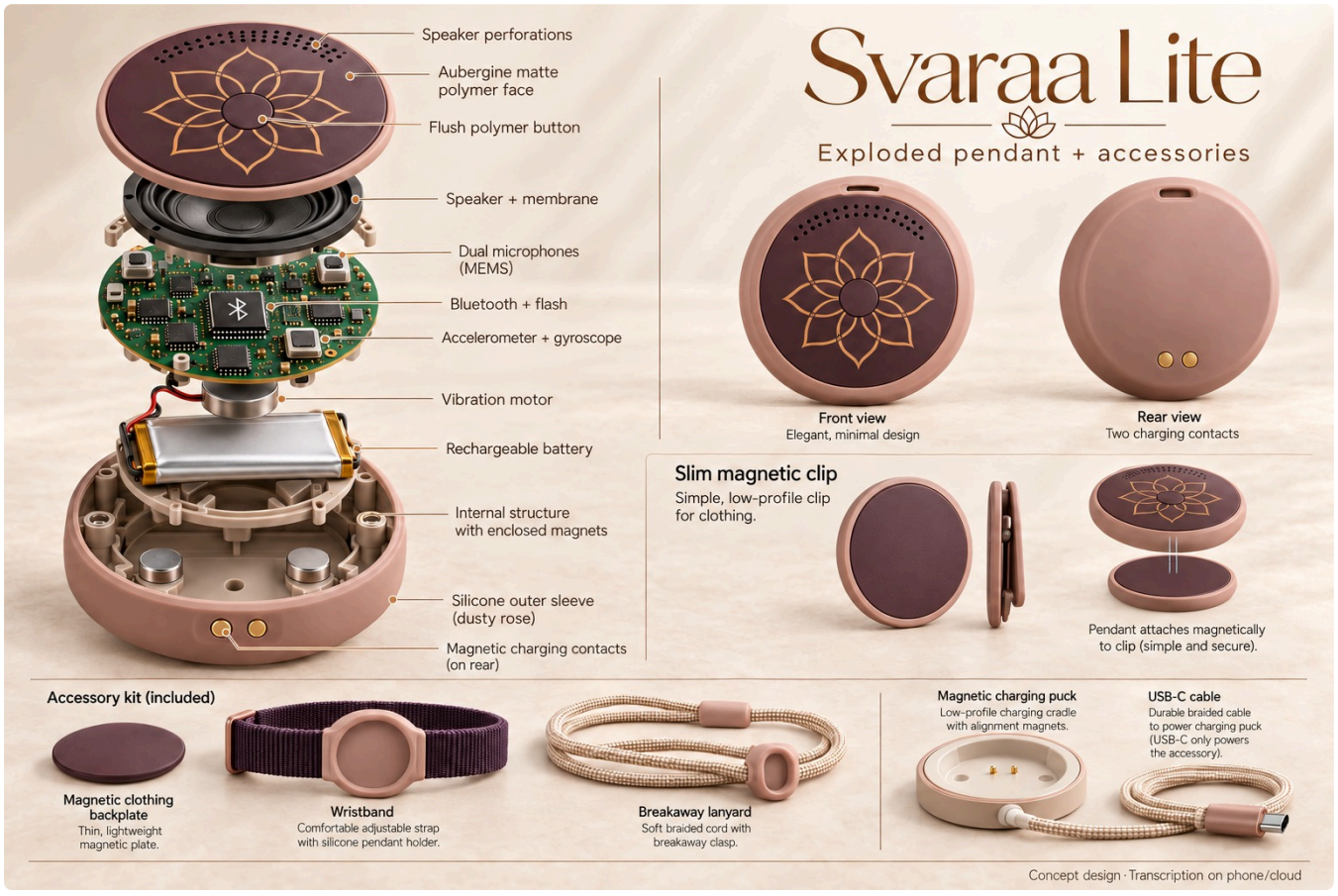
Svaraa Lite Accessory Edition — Hardware Specification

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Summary

Svaraa Lite Accessory Edition (SL1-A) is the Svaraa Lite pendant rebuilt as a sealed, port-free module that snaps into a kit of wear and charging accessories. Electronics, firmware, app and cloud are the Svaraa Lite SL1 v1.0 design; what changes is the enclosure, magnetic attachment and charging through two rear contacts instead of a USB-C port.



Item	Value
Issued by	Svaraa Ventures Inc.
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Product code	SL1-A (pendant + accessory kit, single SKU)
Base platform	Svaraa Lite SL1 v1.0 (23 Sep 2026): same PCB function, firmware, app and cloud

Item	Value
Look	Aubergine matte polymer face with gold lotus, flush centre button, dusty-rose silicone sleeve
Envelope	Ø42.0 × 10.2 mm with sleeve, ≈ 25 g, IP68 goal / IP67 minimum (estimates)
Battery life	≈ 4–5 days typical, ⩾ 2 days heavy use (estimates, as SL1)
Charging	Magnetic pogo-pin puck, powered by any USB-C cable and charger; ≈ 2 h to full
Ways to wear	Breakaway lanyard, slim magnetic clip, magnetic clothing backplate, wristband
BOM target	≈ $43 pendant + ≈ $11.5 accessory kit = ≈ $55 at 10k units (estimate)
Programme	EVT 8–10 weeks from kickoff · MP ≈ 7–8 months; accessory tooling runs in parallel

Like SL1, the Accessory Edition uses only standard, commercially available components and methods and does not include the proprietary platform features of the full Svaraa line. Numbers marked "estimate" are engineering estimates for partner review. Anything not restated here follows the SL1 v1.0 specification.

1 Edition at a glance

The Accessory Edition trades SL1's "any USB-C cable charges it" for a sealed, port-free pendant that can be worn four ways. Polymer and silicone replace the steel bezel and glass face, so it is about 5 g lighter; the attachment magnets make it 0.4 mm thicker.

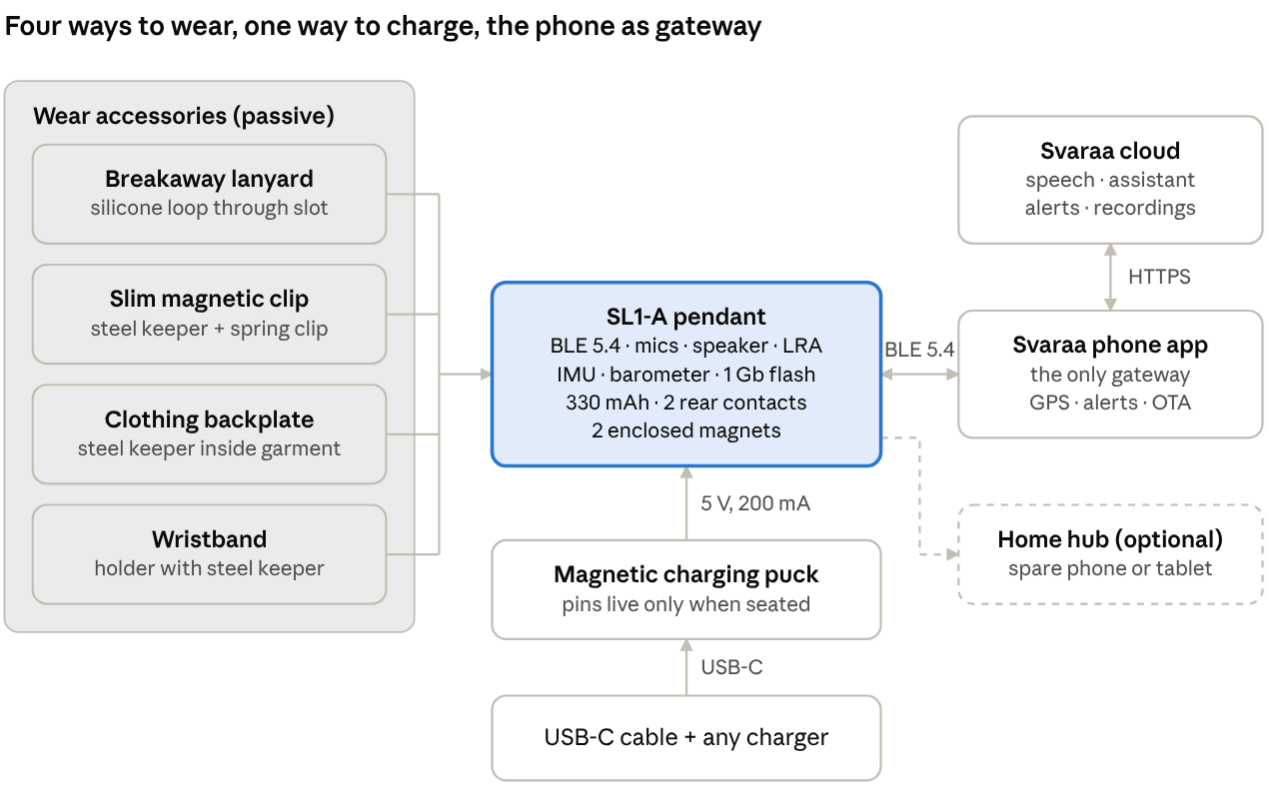
Area	Svaraa Lite (SL1)	Accessory Edition (SL1-A)
Electronics	nRF5340, 2 mics, IMU, barometer, 1 Gb NOR, speaker, LRA	Same, minus USB data and the USB-C receptacle
Enclosure	316L PVD rose-gold bezel, 0.8 mm printed glass face	Glass-filled PC housing, matte PC face with in-mould aubergine and gold lotus, dusty-rose LSR sleeve
Button	Ø6 mm gold PVD cap, 0.3 mm proud	Flush polymer cap with a 0.2 mm tactile rim; same dome and membrane

Area	Svaraa Lite (SL1)	Accessory Edition (SL1-A)
Charging	Waterproof USB-C receptacle at 6 o′clock	Two gold contacts on the rear; magnetic puck powered by USB-C
Sealing	IP67 with the port open	IP68 goal (IP67 minimum); no openings except acoustic membranes
Attachment	Integral bail on the bezel	Two enclosed magnets; slot at 12 o′clock in the sleeve
Ways to wear	Pendant on rudraksha mala	Breakaway lanyard, magnetic clip, clothing backplate, wristband; mala optional through the slot
Size / weight	Ø42 × 9.8 mm, ≈ 30 g	Ø42 × 10.2 mm, ≈ 25 g
Battery	330 mAh, ≈ 4–5 days typical	Same
BOM (10k, estimate)	≈ $50 pendant + cable	≈ $43 pendant + ≈ $11.5 accessory kit
Certification	FCC, ISED, WPC ETA, BIS, CE RED	Same, plus the charging puck as a USB-powered accessory

The wearer gives up one thing: a USB-C cable alone no longer charges the pendant, so a lost puck stops charging until a spare arrives. In return there is no port to collect sweat, talc or paste.

2 System architecture & ways to wear

The pendant is the only part with electronics that talk; the wear accessories are passive, the puck only supplies 5 V, and every internet function still goes through the paired phone or an optional home hub.



SL1-A system · 4 wear accessories, 1 charger, phone gateway

The four wear accessories all attach to the same two magnets (or the 12 o'clock slot for the lanyard), so changing how the pendant is worn never touches its electronics or seal.

3 The seven R's on the Accessory Edition

All seven R's work as in SL1; only Recite, Respond, Remind and Rescue adapt to where the pendant is worn, and the wearer never has to set this up by hand beyond choosing a wear mode once.

R	What changes on SL1-A
01 Recite	Same audio path (Opus over BLE to the pendant speaker). Volume preset per wear mode, because the wrist sits further from the ear than the chest.
02 Respond	Same press-to-talk path. In wrist mode the wearer raises the wrist to speak; setup teaches this with one spoken example.
03 Remind	Same on-pendant schedule and real-time clock, so reminders fire offline. Wrist mode uses a lighter vibration pattern, since the LRA is felt more strongly on bare skin.

R	What changes on SL1-A
04 Rescue	Fall detection uses a profile per wear mode (section 9.1). Wrist mode shows as "reduced fall sensitivity" in the app until it is validated at DVT.
05 Recall	Unchanged: cloud transcription and search.
06 Repository	Unchanged: 1 Gb NOR holds ≈ 17 h of Opus audio, synced over BLE.
07 Riddle	Unchanged: all integrations live in the app and cloud.

4 Electrical architecture & components

The SL1 block diagram carries over unchanged except at the power input: the USB-C receptacle and USB data lines are removed, and two rear contacts feed the nPM1300 through protection that keeps battery voltage off the contacts at all times.

Power input chain: rear contacts (VBUS, GND) → TVS → reverse-polarity P-FET → OVP/eFuse → nPM1300 VBUS. Because the contacts are input-only and float when off the puck, sweat, a coin or a steel keeper across them does nothing, and there is no standing voltage on the skin to drive corrosion.

Function	Reference part	Requirement / notes	Alternates
Charge contacts	2 × Ø2.5 mm insert-moulded pads, phosphor bronze, hard gold ≥ 0.76 µm over nickel	VBUS + GND at 4.0 mm pitch, flush to −0.1 mm, 8 mm below centre toward 6 o'clock; ≥ 10k mating cycles with the puck	Gold-plated stainless pins
Input protection	TVS on VBUS + P-FET reverse protection + TI TPS2595-class eFuse	Survives +20 V and −5 V on the pads; ESD ±8 kV contact; ≤ 0.1 V on the pads in every state including ship mode	Discrete OVP FET
Attachment magnets	2 × Ø10 × 1.5 mm NdFeB, grade N48H (≥ 120 °C), Ni-Cu-Ni plated	Opposite poles facing out, so the puck seats one way only; fully encapsulated in the back cover	N45SH

Function	Reference part	Requirement / notes	Alternates
Programming	SWD test pads on the PCB	Firmware loaded at PCBA test before the back cover closes; afterwards BLE only (MCUmgr/SMP, MCUboot dual slot)	Tag-Connect footprint
Wear sense (optional)	TI TMAG5273 3D Hall sensor, SOT-23-6, footprint only at EVT	Evaluate whether the field shift when a magnet sits on a steel keeper can detect clip and backplate wear automatically	TI DRV5032 Hall switch
Removed from SL1	Waterproof USB-C receptacle, USB D+/D− routing, CC pull-downs	Saves ≈ $1.5 and 0.4 mm of PCB-zone height	—

All other parts (nRF5340, TDK T5838 mics, LSM6DSV16X IMU, BMP581 barometer, 1 Gb QSPI NOR, MAX98357A, 15 × 11 × 3 mm speaker, LRA + DRV2605L, nPM1300, Ø35 × 2.7 mm 330 mAh cell) follow the SL1 component table. The nPM1300 still wakes from ship mode on VBUS, so a new unit wakes the first time it is set on the puck.

5 Mechanical design, magnets & rear contacts

The pendant is a sealed polymer core inside a replaceable silicone sleeve, Ø42.0 × 10.2 mm overall, with the same aubergine face, gold lotus and centre button as SL1 from the front. A polymer housing is RF-transparent, so SL1's bezel RF window is no longer needed.

Layer (front to back)	Thickness
Sleeve lip over the face edge	0.3 mm
Matte PC face, in-mould aubergine + gold lotus, speaker perforations at 12 o'clock	1.0 mm
Front cavity: button dome + membrane, acoustic ducts, speaker/LRA protrusion	1.2 mm
Main PCB zone: Ø36 mm 6-layer HDI, speaker and LRA in cutouts, no USB-C	3.0 mm
Battery: Ø35 × 2.7 mm round Li-Po, 330 mAh	2.7 mm
Back cover with magnet pockets, insert-moulded contacts, O-ring	2.0 mm

Layer (front to back)	Thickness
Total	10.2 mm (10.5 design-to)

Part	Specification
Core housing	PC + 10 % glass fibre, Ø40.0 mm, 1.0 mm wall; mic ports at 10 and 2 o′clock behind ePTFE membranes
Face	1.0 mm matte PC with in-mould decoration; arc of Ø0.6 mm speaker holes at 12 o′clock behind acoustic mesh and ePTFE; centre aperture for the button
Centre button	Ø6 mm aubergine polymer cap with a printed gold ring, flush with the face, 0.2 mm raised rim so it can be found by touch; same dome + bonded membrane as SL1 (1.8–2.5 N, 0.25 mm travel, 200k cycles)
Back cover	PC + GF in dusty rose; two magnet pockets with 0.3 mm skin; two insert-moulded contacts; O-ring sealed, three screws hidden under the sleeve
Sleeve	Liquid silicone rubber, Shore 50A, dusty rose, 0.8 mm wall; 8 × 3 mm slot at 12 o′clock for the lanyard or mala loop; undercut lip, pull-out force ≥ 30 N; replaceable, colour variants possible
Sealing	IP68 goal (1.5 m, 30 min), IP67 minimum; the only openings are the speaker holes and mic ports, all behind membranes
Mass	≈ 25 g pendant with sleeve (estimate)

Magnetic attachment requirements (all to be measured at EVT):

- Pull-off from the clip or backplate through 2 mm of fabric: ≥ 6 N; with no fabric: ≥ 10 N.
- Slide (shear) resistance through 2 mm of fabric: ≥ 2 N, about 8 times the pendant′s weight.
- On the puck, opposite poles key the orientation: a pendant set down rotated 180° is pushed away and cannot touch the pins.
- No magnet comes loose after 1.5 m drops onto concrete and a 50 N push-out test on each pocket.
- Stray-field limits for wearers with cardiac implants are in section 9.2.

6 Accessory kit (included)

Every accessory is passive except the charging puck. Each one the pendant snaps onto (clip, backplate, wristband holder) carries a 0.5 mm ferritic stainless (430) keeper plate, which gives the magnets something to grip and shunts their field away from the body.

Accessory	Purpose	Specification	BOM (USD, 10k, est.)
Magnetic charging puck	Nightly charging, one-handed	Ø48 × 8 mm cream PC/ABS cradle with a shallow recess matching the pendant back; 2 gold pogo pins (1.0 mm travel, 0.5–1.0 N each); 2 alignment magnets with opposite poles; USB-C receptacle on the back edge (5 V sink, 5.1 kΩ CC pull-downs, no data); TVS + eFuse limited to 400 mA; a Hall switch powers the pins only when a pendant is seated; seating pull 2–4 N; ≥ 80 g with silicone feet so the pendant lifts off without holding the puck	4.0
USB-C cable	Powers the puck only	1 m braided, cream, USB-C to USB-C, 3 A; any USB-C charger or a USB-A to C cable also works	1.2
Slim magnetic clip	Waistband, pocket, belt or bag strap	Ø38 mm disc, ≤ 6 mm thick, aubergine soft-touch over the steel keeper; stainless spring clip on the back gripping ≥ 8 N on 3 mm fabric	1.2
Magnetic clothing backplate	Wear anywhere on a shirt, kurta or blouse	Ø36 × 1.2 mm steel keeper overmoulded in aubergine TPU, rounded edge; worn inside the garment behind the pendant; fabrics up to 3 mm; larger than the 31.7 mm small-parts cylinder	0.5

Accessory	Purpose	Specification	BOM (USD, 10k, est.)
Wristband	Wrist wear	20 mm woven polyester strap, hook-and-loop adjustment for 140–210 mm wrists (easier than a buckle for arthritic hands); LSR holder with a closed back and steel keeper; pendant retention ≥ 20 N	1.8
Breakaway lanyard	Pendant around the neck	Ø3 mm braided polyester cord, 60–80 cm via a silicone slider bead; silicone loop through the sleeve slot; breakaway clasp at the nape releasing at 15–25 N	1.3
Kit packaging	Retail box with formed tray	One cavity per accessory, quick-start card with pictures only	1.5
Total kit	—	—	≈ 11.5

The release forces are ordered on purpose: the lanyard clasp lets go (15–25 N) before the sleeve slips off the pendant (≥ 30 N), and the sleeve slot is specified to survive ≥ 50 N without tearing. Skin-contact materials (sleeve, strap, holder, backplate) are tested to ISO 10993-5 and -10. A rudraksha mala ending in the same silicone loop can be sold separately to keep the SL1 look.

7 Power, contact charging & battery life

Battery life is the same as SL1, ≈ 5 days typical when new, because the electronics and the 330 mAh cell are unchanged; wrist wear adds an estimated 1–2 mAh/day from extra motion wake-ups.

Use profile	mAh/day	New (297 mAh usable)	End of life (80 %)
Light (few questions, no recording)	≈ 35	≈ 8 days	≈ 6.5 days
Typical (SL1 daily profile)	≈ 60	≈ 5 days	≈ 4 days

Use profile	mAh/day	New (297 mAh usable)	End of life (80 %)
Heavy (wake word 16 h + continuous recording 16 h)	≈ 155	≈ 1.9 days	≈ 1.5 days

All figures are engineering estimates from datasheet currents; EVT replaces them with measured values per mode and per wear mode.

Charging on the puck:

- 200 mA CC/CV to full in ≈ 2 h; JEITA derated above 40 °C and suspended above 45 °C, as SL1.
- Seating confirms itself out loud: a tick and "Charging, 40 percent". No announcement within 3 s means the contacts are not touching, so the wearer reseats it.
- Ship mode ⩽ 2 µA for retail storage; a new unit wakes the first time it is set on the puck.
- Contact resistance (pad + pogo pin) ⩽ 150 mΩ, so the drop at 200 mA stays under 30 mV.
- Puck standby draw from USB ⩽ 1 mW with no pendant seated (Hall switch and eFuse only).
- Low-battery behaviour is unchanged: "Please charge me soon" once a day at 15 %, essentials only at 5 %, family warned at 20 %.

8 User interface without lights

Button gestures, vibration patterns and voice prompts are the SL1 set; the Accessory Edition adds a wear-mode setting and a few spoken events for the puck and the clip.

Button gestures are unchanged: single press to talk or acknowledge, hold 3 s for SOS, double press for status, triple press for microphones off/on. The service restart becomes "hold 10 s while on the puck". The status sentence now names the wear mode: "Svaraa is ready, battery 70 percent, on your wrist, connected to your phone."

Event	Feel / hear
Set on the puck	Tick + "Charging, 40 percent"
Lifted off the puck when full	Tick + "Fully charged"
Wear mode changed	One pulse + "I'm on your wrist now" (or clip, chest)
Came off the clip or backplate	Two quick pulses + "I've come loose" (only if wear sensing passes EVT, section 4)

Wear mode is set once, in the app by the wearer or family, or by voice ("Svaraa, I'm wearing you on my wrist"). It sets the volume preset, vibration strength and fall-detection profile. The default is chest wear (lanyard, clip or backplate behave the same).

Finding the button: the flush cap is findable by its 0.2 mm raised rim, a smoother texture than the matte face, and the printed gold ring for low-vision wearers. The DVT home trial checks that wearers find it without looking at least 19 times in 20.

9 Safety: falls, magnets & breakaway

The SL1 fall and SOS method stays; what is new is a fall profile per wear mode and a hard limit on the magnetic field reaching the body, because many elderly wearers have pacemakers or ICDs.

9.1 Fall and SOS by wear mode

Wear mode	Fall profile	Notes
Lanyard (chest, swinging)	SL1 method unchanged: free-fall ≥ 150 ms then impact ≥ 2.5 g; gyroscope rotation ≥ 200 °/s and tilt ≥ 60°; 20 s stillness	SL1 thresholds were set for a swinging pendant
Clip or backplate (fixed to the torso)	Same method; tilt measures the torso directly, so thresholds may be tightened from DVT data	Fixed mounting gives cleaner signals and fewer false alarms
Wrist	Wrist-tuned thresholds that reject arm swings and gestures; longer stillness window set from DVT data	Shown as "reduced fall sensitivity" until validated on a wrist dataset
On the puck	Fall detection paused	Hold-3-s SOS still works

Hold-3-s SOS, the 30 s "Did you fall?" countdown and the out-of-range loud alarm are the same in every mode. If the lanyard breaks away and the pendant hits the floor, the normal countdown runs; the wearer cancels with a press or "I'm fine".

9.2 Magnets and cardiac implants

Pacemakers and ICDs are tested to tolerate static fields only up to 1 mT (10 gauss) for temporary malfunction (ISO 14117:2019, clause 4.6). Our estimate is that two bare Ø10 × 1.5 mm N48H magnets still give ≈ 1 mT at about 30 mm, so a pendant clipped over a left-chest

implant needs a limit.

- Requirement, bare pendant (lanyard): ⩽ 1 mT at 30 mm from the back surface.
- Requirement, on a keeper (clip, backplate, wristband): ⩽ 1 mT at 10 mm behind the keeper, which the steel plate and the opposite-pole magnet pair should make achievable.
- Charging puck: ⩽ 1 mT at 30 mm from any surface with a pendant seated.
- All measured with a calibrated gaussmeter at EVT and DVT; a cardiac-device consultant reviews the data before launch.
- The app asks at setup whether the wearer has a pacemaker or ICD; if yes it recommends wrist wear, never a clip on the left chest, and puts the backplate on the pendant's back as a shield in lanyard wear.
- Leaflet and box carry an implant warning, and the final separation distance is set from the measured field.

9.3 Small parts and breakaway

- Magnets are fully encapsulated; no accessory contains a loose magnet, and the backplate (Ø36 mm) is larger than the small-parts cylinder. Two swallowed magnets are a serious hazard for grandchildren in the home, so drop, push-out and bite tests cover every magnet and keeper.
- The lanyard releases at 15–25 N, so it cannot become a strangulation hazard if it catches on a bed rail, walker or door handle.

Location (from the phone's GPS), lost-pendant finding and integrations are unchanged from SL1.

10 Firmware & software integration

Firmware, app and cloud are the SL1 stack (Zephyr on nRF5340, MCUboot, standard BLE services plus the encrypted Svaraa Lite control service); the Accessory Edition adds a wear-mode setting and loses the USB path. The charging puck has no firmware at all.

Item	Change from SL1
Wear mode (new characteristic)	Read/write: chest, wrist; selects volume preset, vibration strength and fall-profile parameters; stored on the pendant so it survives without the phone

Item	Change from SL1
Fall profiles	Thresholds per wear mode held in settings and updatable over the air, so DVT tuning ships without a hardware change
Events (notify)	"Charger attached" now comes from the rear contacts; new "wear mode changed" and, if wear sensing passes EVT, "came off keeper"
Firmware updates	BLE only (MCUmgr/SMP); OTA runs only while on the puck with battery ≥ 50 %
Factory programming	SWD at PCBA test; final assembled units are checked over BLE and through the contacts
App setup	Adds a picture-led wear-mode choice and the cardiac-implant question (section 9.2)

The SL1 reliability rules are unchanged: reminders, fall detection and the SOS countdown never depend on the phone, events are stored until the app confirms receipt, and family is warned if the pendant has not connected for 2 h during the day or has not been charged for 3 days.

11 Certification, testing, cost, timeline & risks

The Accessory Edition reuses every SL1 radio approval path and adds the charging puck as a USB-powered accessory with no radio; the pendant plus kit lands at ≈ $55 at 10k units (estimate).

Certifications: SL1 set unchanged for the pendant (FCC Part 15C + 15B, ISED RSS-247, WPC ETA, BIS CRS, CE RED incl. EN 18031, UKCA, Bluetooth SIG, UN 38.3, IEC 62133-2, RoHS, REACH, WEEE). The puck adds FCC Part 15B, CE EMC and IEC 62368-1. Skin-contact parts are tested to ISO 10993-5 and -10.

Production tests (100 %): the SL1 list without the USB data test, plus charging through the rear contacts, contact resistance, magnet presence and polarity with a Hall probe, sleeve fit, and pressure-decay leak test. Every puck is checked for dead pins with no pendant, correct seating detection and the eFuse trip.

Qualification: the SL1 list (IP67/IP68, 1.5 m drop, −10 to +45 °C, 85 % RH 40 °C 96 h, 200k button presses, 500 battery cycles, ESD ±8 kV) plus 10k puck mating cycles; artificial sweat on the contacts powered and unpowered; magnet pull and shear at −10 and +45 °C;

magnet push-out and bite tests; 5k clip and hook-and-loop cycles; lanyard release force over 1,000 cycles; stain tests on the dusty-rose sleeve with sweat, turmeric, kumkum and sandalwood paste; static-field mapping per section 9.2.

BOM item (10k units, estimate)	USD
nRF5340 + crystals + antenna + RF passives	9.0
2 mics, IMU (accel + gyro), barometer	6.0
Amp, speaker, LRA + driver	5.0
NOR flash, PMIC, input protection	5.5
Battery 330 mAh round	3.0
PCB 6-layer HDI + assembly + test	8.5
Polymer housing, face, button, back cover, membranes	4.5
LSR sleeve	0.8
Magnets + contact pads	1.0
Pendant	≈ 43
Accessory kit (section 6)	≈ 11.5
Pendant + kit	≈ 55 (50–65)
NRE: tooling (incl. 5 accessory tools) + design + certification	≈ 130–190k

Timeline, from kickoff:

1. EVT, 8–10 weeks: 50 units, all functions, power and magnetic field measured, first accessory soft tools.
2. DVT, 8 weeks: 200 units with full kits; final mechanics and reliability; 20-elder home trial across all four wear modes; wrist fall dataset; pre-scan.
3. PVT, 6 weeks: 1k units; certifications complete; yield ≥ 95 %.
4. MP: ≈ 7–8 months from kickoff.

Risk	Mitigation
Lost or broken puck stops charging	Puck runs on any USB-C cable and charger; spare pucks sold; app warns family at 20 % and after 3 days uncharged; option of two pucks in the kit (+≈ $4)
Magnetic field near a pacemaker or ICD	Field limits and keeper plates (9.2); implant question at setup; wrist-wear recommendation; consultant review; labelling
Pendant knocked off the clip or backplate and lost	Pull and shear requirements; optional "I've come loose" alert; Find My / Find Hub as SL1
Contact corrosion from sweat, talc or paste	Input-only floating contacts; hard gold ≥ 0.76 µm; powered and unpowered sweat tests; dry-before-charging prompt
Wrist fall detection weaker than chest	Wrist profile; "reduced fall sensitivity" label until a DVT dataset validates it
Flush button hard to find	Raised rim, texture and gold ring; findability checked in the DVT home trial
Light sleeve stains	Stain tests; replaceable sleeve; darker colour options
Third-party patents on magnetic attachment and contact charging	Freedom-to-operate review before launch, alongside the SL1 review of fall detection and recording